CSE_2323

Binary Codes

Digital Logic Design

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BINARY CODES

Digital Codes or Binary Codes:

A group of binary digits is known as a binary code or a digital code.

Numeric Code:

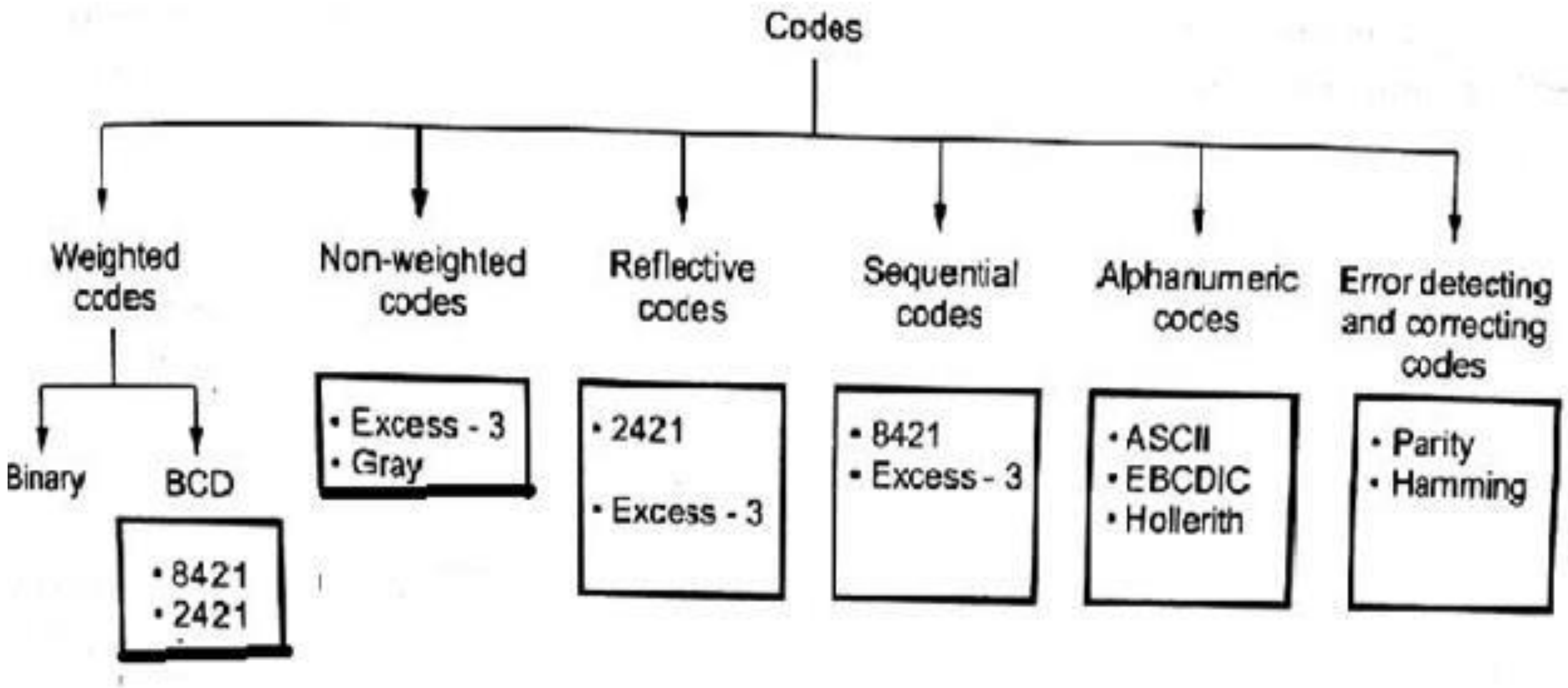
Digital code represented by the number is known as a numeric code.

Classification:

The codes are broadly classified into

1. Weighted code
2. Non-weighted code
3. Reflective code
4. Sequential code
5. Error detecting and correcting codes

Classification



Weighted codes

the position of each number represent a specific weight.
In these codes each decimal digit is represented by a group of four bits.called Nibble

(Examples:8421, 2421 Code)

each digit is assigned a specific weight according to its position. For example, in 8421/BCD code, 1001 the weights of 1, 1, 0, 1 (from left to right) are 8, 4, 2 and 1 respectively.

BCD and 2421 Code

Decimal Equivalent	BCD Code				2421 Code			
	D_4	D_3	D_2	D_1	Y_4	Y_3	Y_2	Y_1
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	0
3	0	0	1	1	0	0	1	1
4	0	1	0	0	0	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	1	0	0
7	0	1	1	1	1	1	0	1
8	1	0	0	0	1	1	1	0
9	1	0	0	1	1	1	1	1

8421 or BCD code

BCD stands for Binary-Coded Decimal.

A BCD number is a four-bit binary group that represents one of the ten decimal digits 0 through 9.

Decimal Equivalent	BCD Code D_4	D_3	D_2	D_1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

Example:

Decimal number 4926 4 9 2 6

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

8421 BCD coded number $\boxed{0100\ \ 1001\ \ 0010\ \ 0110}$

8421 or BCD code

- (BCD) is a type of binary code used to represent a given decimal number in an equivalent binary form.
- The BCD equivalent of a decimal number is written by replacing each decimal digit in the integer and fractional parts with its four-bit binary equivalent.
- The BCD code described above is more precisely known as the 8421 BCD code .

As an example, the BCD equivalent of $(23.15)_{10}$ is written as $(0010\ 0011.0001\ 0101)_{BCD}$

BCD Addition

Addition of BCD (8421) is performed by adding two digits of binary, starting from least significant digit. In case if the result is an illegal code (greater than 9) or if there is a carry out of one then add 0110(6) and add the resulting carry to the next most significant.

For example:-

6	0	1	1	0	←	BCD for 6
+ 3	0	0	1	1	←	BCD for 3
9	1	0	0	1	←	BCD for 9

6	0	1	1	0	←	BCD for 6
+ 8	1	0	0	0	←	BCD for 8
14	1	1	1	0	←	Invalid BCD number

6		0	1	1	0	←	BCD for 6
+ 8		1	0	0	0	←	BCD for 8
14		1	1	1	0	←	Invalid BCD number
	+	0	1	1	0	←	Add 6 for correction

$$\underbrace{0\ 0\ 0\ 1}_{1}\quad \underbrace{0\ 1\ 0\ 0}_{4} \quad \leftarrow \text{BCD for 14}$$

BCD Addition

$$\begin{array}{rcc} & 0100 & 0011 \\ + & 0011 & 0101 \\ \hline & 0111 & 1000 \end{array}$$

$$\begin{array}{r} 43 \\ +\ 35 \\ \hline 78 \end{array}$$

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$$\begin{array}{cccc} & & 0111 & 0101 \\ & + & 0011 & 0101 \\ \hline & & \color{red}{1010} & \color{red}{1010} \\ & & +0110 & +0110 \\ \hline & \underbrace{0001}_{1} & \underbrace{0001}_{1} & \underbrace{0000}_{0} \end{array}$$

Both left and right BCD numbers are invalid. So we would add 6 to both the BCD numbers.

$$\begin{array}{r} 75 \\ +\ 35 \\ \hline 110 \end{array}$$

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BCD Addition

$(47+65)=?$

1 Marks.

BCD Addition

For example:-

Add 8765 with 3943 using BCD addition.

Solution:-

		111	111	1	111
		1000	0111	0110	0101
+		0011	1001	0100	0011
		1 1100	0000	1010	1000
Correction →		0110	0110	0110	0000
	0001	0010	0111	0000	1000
	1	2	7	0	8

Non-weighted code

Non-weighted codes:

The non-weighted codes are not positionally weighted. In other words,

codes that are not assigned with any weight to each digit position.

Example: Excess-3(XS-3) and Gray Codes.

Excess-3 code

Excess-3 codes are Non-weighted and can be obtained by adding 3 to each decimal digit then It can be represented by using 4 bit binary number for each digit.

Non-weighted code

An Excess-3 equivalent of a given binary number is obtained using the following steps:

- Find the decimal equivalent of the given binary number.
- Add +3 to each digit of decimal number.
- Convert the newly obtained decimal number back to binary number to get required excess-3 equivalent.

- These are following excess-3 codes for decimal digits –

Decimal	BCD 8421	Excess – 3 code BCD + 0011
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

Gray Code

Gray code:

Gray code has a property that two successive numbers differ in only one bit

Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

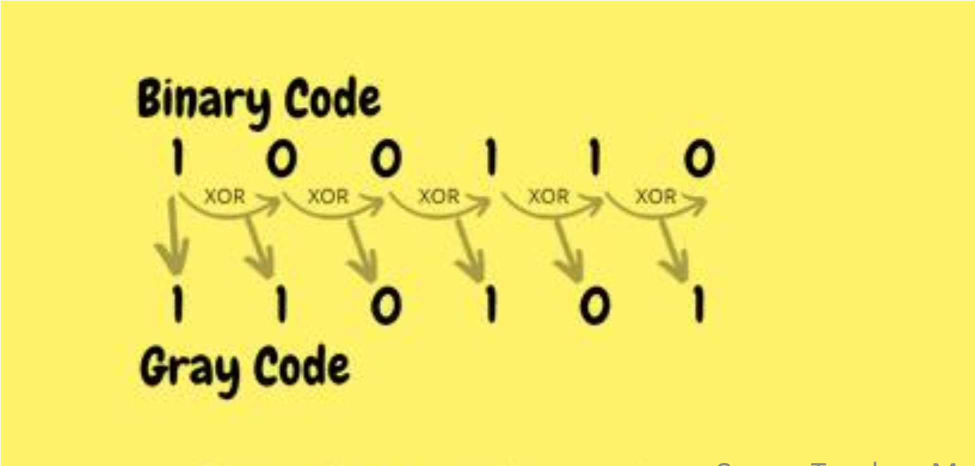
Applications:

It is used in analog to digital conversion, input / output devices.
It is used to reduce errors that occur in data transmission.

Binary to Gray conversion

- The MSB of the gray code is the MSB of the binary number.
- Perform XOR operation between the MSB and the second significant bit of the binary number.
- XOR the second and third significant bits of the binary, the result is the third significant bit of the gray code.
- Repeat the process till the end of the LSB of the binary number

Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

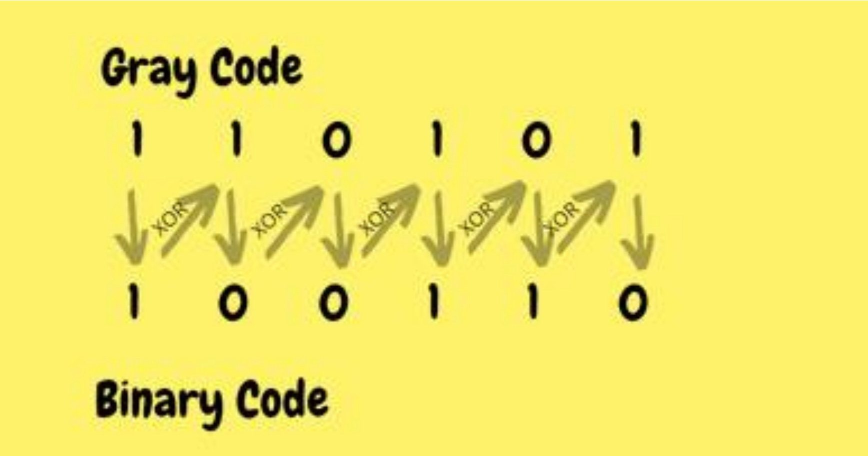


Binary code to gray code conversion

Gray to Binary conversion

- The MSB of the binary is same as the MSB of gray code.
- Perform the XOR operation between the MSB of the binary and the second significant bit of gray code the result is the second significant bit of binar
- Perform the XOR operation between the second significant bit of binary and the thir significant bit of gray code the result is the third significant bit of binary.
- Repeat the process until all the gray code bits are XOR.

Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000



Gray code to binary code conversion

Reflective and sequential code

A code is said to be reflective when the code for 9 is the complement for the code 0
8 is the complement for the code 1
7 is the complement for the code 2
6 is the complement for the code 3

Decimal digit	2	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	0	1
8	1	1	1	0
9	1	1	1	1

.

Sequential codes:

A code is said to be sequential when each succeeding code is binary number greater than its preceding code.

Eg: 8421 code and Excess-3 code

Self complimenting code

For self rcomplimenting code must to be fulfilled this condition

$$(W_4 + W_3 + W_2 + W_1 = 9)$$

For 8421 code =8+4+2+1=15 (8421 is not self complimenting code)

For 4221 code =4+2+2+1=9 (4221 is self complimenting code)

For 2421 code =2+4+2+1=9 (2421 is self-complimenting code

For 5421 code =5+4+2+1=12((5421 is not self complimenting code)

Decimal	8421 BCD code	4221 BCD code	5421 BCD code
0	0000	0000	0000
1	0001	0001	0001
2	0010	0010	0010
3	0011	0011	0011
4	0100	1000	0100
5	0101	0111	1000
6	0110	1100	1001
7	0111	1101	1010
8	1000	1110	1011
9	1001	1111	1100

Parity codes

used for the purpose of detecting errors during the transmission of binary information. The parity code is a bit that is included with the binary data to be transmitted.

Based on the number of 1's in the transmitted data, the parity code is of two types.

1.

1. Even parity code
2. Odd parity code

Parity codes

Even parity code:

If the total number of I bits in the word including parity bit is Even, then such a parity code is said to be Even parity code.

Odd parity code:

If the total number of I bits in the word including parity bit is odd, then such a parity code is said to be odd parity code.

The following table shows the even and odd parity bits for 4 bit data word

Data	Even parity	Odd parity
0 0 0 0	0	0
0 0 0 1	1	0
0 0 1 0	1	0
0 0 1 1	0	1
0 1 0 0	1	0
0 1 0 1	0	1
0 1 1 0	0	1
0 1 1 1	1	0
1 0 0 0	1	0
1 0 0 1	0	1

The simple parity will not detect two errors within the same word. To detect and correct error in the word Hamming code is used.

Hamming code

Hamming code

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver. It is technique developed by R.W. Hamming for error correction.

Redundant bits/parity bits

Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the data transfer.

Generation of hamming code

Selecting the number of parity bits:

The number of parity bits to be chosen depends on word length.the parity bits are even parity bits
Let **n** be the number of information or data bits, then the number of parity bits P is determined from the following formula,

$$2^P \geq n + P + 1$$

Example:

if 4-bit information is to be transmitted, then n=4. The number of parity bits is determined by the trial and error method

Let P=2, we get,

$$2^2 \geq 4 + 2 + 1$$

The above equation implies 4 not greater than or equal to 7. So let's choose another value of P=3.

$$2^3 \geq 4 + 3 + 1$$

Now, the equation satisfies the condition. So number of parity bits, P=3. The bit designation is

7	6	5	4	3	2	1
D4	D3	D2	**P3**	D1	**P2**	**P1**

Example:Generation of hamming code

Problem:

Data bits 1011 must be transmitted. Construct the even parity, seven bit Hamming code for this data

D4	D3	D2	**P3**	D1	**P2**	**P1**
1	0	1		1		

P1 checks for even parity of bit positions 1,3,5 and 7. For even parity **P1 must be 1**

D4	D3	D2	**P3**	D1	**P2**	**P1**
1	0	1		1		**1**

P2 checks for even parity of bit positions 2,3,6 and 7. For even parity **P2 must be 0**

D4	D3	D2	**P3**	D1	**P2**	**P1**
1	0	1		1	**0**	**1**

P3 checks for even parity of bit positions 4,5,6 and 7. For even parity **P3 must be 0**

D4	D3	D2	**P3**	D1	**P2**	**P1**
1	0	1	**0**	1	**0**	**1**

1.To determine the value of P1, the sequence as (1,3,5,7).As the value of location 3 is 1, location 5 is 1, and location 7 is 1. So a number of one is the sequence is three, which is an odd number. So to satisfy even parity, the value of P1 must be 1.

2.To determine the value of P2, first,The sequence is (2,3,6,7). the digits from the location number we get (P2,1,0,1). So a number of ones are two. Which is an even number. So to satisfy even parity, the value of P2 must be zero.

1.To determine the value of P3, first, The sequence is (4,5,6,7). the digits from the location number we get, (P3, 1,0,1). So the number of one's is two. Which is an even number. So to satisfy even parity, the value of P3 must be zero.

2.Finally, after obtaining the values of P1, P2, and P3, the complete hamming code along with the data message is given as

Hamming Code Detection with Even Parity

A hamming code 1110101 is received at the receiver end.what are the correct data bits.

7	6	5	4	3	2	1
D4	D3	D2	P3	D1	P2	**P1**
1	1	1	0	1	0	1

the values of locations 1,3,5, and 7 becomes (1,1,1,1,). So a number of 1 is the sequence is four.which is an even number . So P1 parity bit is right

the values of locations 2,3,6, and 7 becomes (0,1,1,1,). So a number of 1 is the sequence is three.which is odd number . So P2 parity bit is wrong

the values of locations 4,5,6, and 7 becomes (0,1,1,1,). So a number of 1 is the sequence is three.which is odd number . So P3 parity bit is wrong

For wrong parity palce 1.and write parity place 0

p3	P2	P1
1	1	0

Decimal value of 110 is 6 .it indicates that there is error occurred in bit number 6.so bit number 6 will be 0.

so correct data is 1 of hamming code will be 1011


THE
END